

Statistical Inference and Learning [CM0471]

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Exam Assignment Report

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Predicting Excessive Enforcement Patterns in Police Behavior After Vehicles Stops in New York

Methodology

This project studies arrest outcomes following NYPD vehicle stops in New York City. We study **whether the observed stop resulted in an arrest and which characteristics help explain or predict that outcome**. The statistical target is therefore $P(\text{arrested} = 1 \mid X)$, where X contains the predictor’s information.

Data Origin: The data comes from the *NYPD Vehicle Stop Reports*, published by the City of New York. The raw file used for the project contains 2,488,293 stop records and 19 attributes covering time, location, officer command, vehicle type, motorist demographics and enforcement outcomes.

Data Context and Problems: The data covers only about three years, it provides no complete description of the reason for the stop, and it contains partially missing demographic information. Coordinates are also incomplete or rounded in a non negligible fraction of observations.

Data Preparation: The main objective was to retain information that could help us generalize arrest-related variables, while removing identifiers, overfitting predictors and features that could increase model complexity without adding interpretable signal. Here, we sum up what we did:

Block	Main operation	Final representation
Target	retain arrest only	<code>arrested</code> $\in \{0, 1\}$
Time	parse and decompose	year, month, weekday, hour
Demographics	clean and group sparse values	age, age bracket, sex, ethnicity
Geography	aggregate high cardinality command	area, boro, coordinates for EDA
Vehicle	remove commercial and TLC groups	2 wheels, 4 wheels

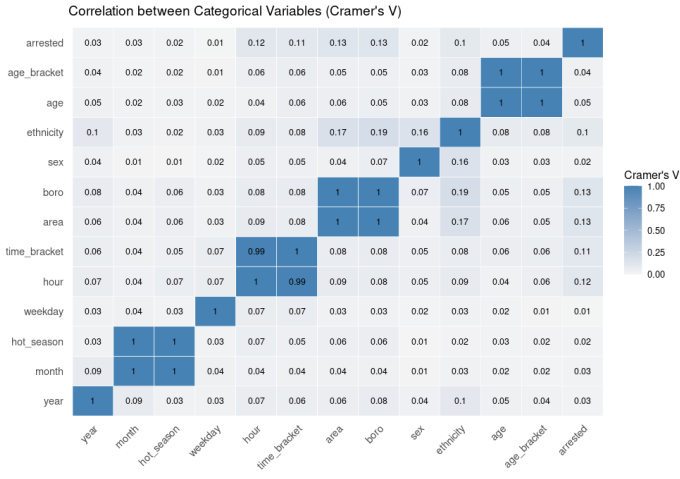
In particular, we must mention that missing demographic values were preserved through explicit Unknown levels (when appropriate), because we considered that the complete deletion would alter the empirical population instead of removing noise. Furthermore, the bracketing of commands was done based on a report published by the NYPD, and the categories of vehicles that we retain are either any vehicles with 2 wheels or private vehicles with 4 wheels. These two macro categories originated 2 separate datasets. After cleaning, we understood that the data available was very difficult to deal with. Most useful predictors are categorical and the main numerical variables, age and hour, are not naturally represented by a simple linear effect. At that point, we chose to go for parsimonious and interpretable models.

Splitting the Analysis: The cleaned datasets (from the data preparation) are **analysed as two separate populations with the same response and the same research focus**. This split is mainly justified by the strong differences in size, arrest association, age composition and temporal/geographical distributions.

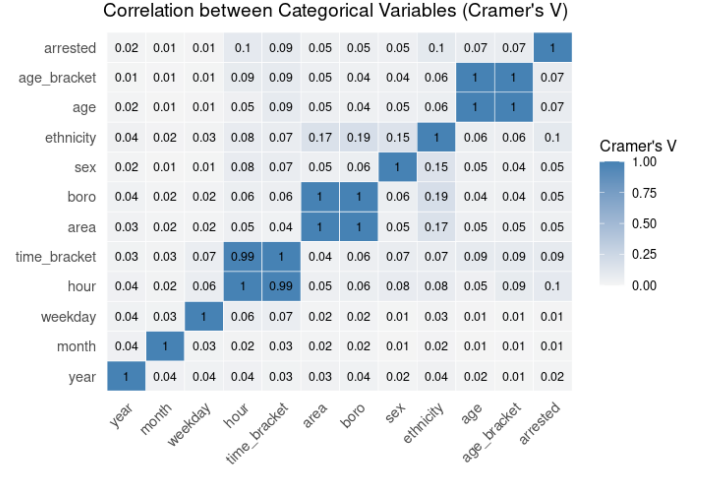
Exploratory Data Analysis and Goal: The EDA is used to identify the structure that should be carried into the models. Its main purpose is to determine which predictors deserve attention, where the two datasets agree or disagree, and which relationships do not show linearity.

Univariate Analysis: First of all we have huge difference in scale and class prevalence. The two wheeled dataset contains 189,610 stops, with 10,560 arrests, corresponding to approximately 5.6%. The four wheeled dataset contains 2,157,445 stops, with 70,600 arrests, approximately 3.3%. **Arrest is, thus, a rare outcome in both populations** and especially rare for four wheeled vehicles. Secondly, the distribution of the predictors also highlight how the two dataset differ. **Two wheeled stops are concentrated among younger motorists** and exhibit pronounced seasonal variation, with much higher volumes during warmer months and strong growth from 2023 to 2025. **Four wheeled stops have a broader age distribution and a much more stable annual and seasonal profile**. Geographic distribution of stops is also more even for four wheeled vehicles. These differences justify the separate analyses but, at the same time, both datasets share the same difficulty: a rare binary target must be explained using mostly categorical or discrete predictors.

Bivariate Analysis: We first use Cramer's V to summarize pairwise association among categorical predictors. For two wheeled vehicles, the clearest pairwise associations with arrest are geographical location ($V \approx 0.13$ for area and boro), hour or time bracket ($V \approx 0.11$) and ethnicity ($V \approx 0.10$). Sex ($V \approx 0.02$) and age or age bracket ($V \approx 0.04$) are weak when considered alone. For four wheeled vehicles, ethnicity remains around $V \approx 0.10$, hour around 0.09, age bracket around 0.07, sex around 0.05 and geography around 0.05 to 0.06. The matrices also show that **predictor to predictor associations are generally modest**. The largest recurring structure is between ethnicity and geography, approximately 0.17 to 0.19. Pearson correlations among the raw numerical variables are similarly weak. This is useful for diagnostics purposes but it also suggests that simple linear target-predictor relationships do not suffice to explain the target. The bivariate results therefore focus the remainder of the analysis on five axes: boro, hour, sex, ethnicity and age or age_bracket.



(a) Two wheeled vehicles



(b) Four wheeled vehicles

Figure 1.1: Cramér's V association matrices for the two analytical populations.

Geography: The Bronx has the highest raw arrest rate in both populations. For two wheeled stops, borough level rates range from approximately 3.2% in Manhattan to 12.1% in the Bronx. For four wheeled stops, we range from about 2.5% in Queens to 4.8% in the Bronx.

Hour: It produces the most consistent raw pattern across the two populations. **Arrest rates are lowest during the morning and increase during late night and early morning hours.** Around 07:00 the raw rate is approximately 3.2% for two wheeled vehicles and 0.6% for four wheeled vehicles, while around 03:00 it rises to about 10.9% and 5.3%. The shape is **clearly non linear** and motivates treating hour as more than a standard numerical predictor.

Sex: Male motorists have higher raw arrest rates in both datasets. The difference is approximately 5.7% versus 2.7% for two wheeled stops and 3.70% versus 1.51% for four wheeled stops.

Ethnicity: Reported ethnicity shows the strongest demographic contrast. In the two wheeled population, Hispanic and Black motorists have arrest rates of 7.24% and 6.86%, compared with 3.23% for White motorists. In the four wheeled population the corresponding rates are 4.66%, 5.02% and 1.15%.

Age: Age provides the clearest example of disagreement between the populations. Two wheeled arrest rates are highest among the youngest groups and do not follow the same ordering observed for cars. In the four wheeled population, the 25–34 and 18–24 groups have the highest rates, while arrest frequency declines strongly at older ages.

Statistical Modelling and Results

Modelling Strategy: Our Model choice follows the same principle of parsimony. Logistic Regression provides an **interpretable baseline**, **LASSO tests whether its main signals survive shrinkage**, **GAM adds controlled non linear flexibility**, and **Naive Bayes checks the problem from a different probabilistic perspective**. When a tuning objective is required, we often prefer ROC-AUC to likelihood deviance because AUC evaluates ranking over all thresholds and is not dominated by the majority class in the same way as raw classification accuracy. It also avoids privileging one classification direction during tuning. For thresholded evaluation, the operating point is selected on the training data with Youden’s criterion, which gives sensitivity and specificity equal weight. This is consistent with the project assumption that missing an arrest and incorrectly labeling a non-arrest are both relevant errors.

Logistic Regression: The fitted models reproduce the main EDA patterns while clarifying which differences remain after adjustment. **Time of day is the strongest common signal**. Relative to midnight, the estimated odds at 07:00 are $OR = 0.035$ for two wheeled vehicles and $OR = 0.160$ for four wheeled vehicles. Thus, conditional arrest odds are approximately 96.5% and 84.0% lower at 07:00 than at midnight, holding the other predictors fixed. The **important result is the broad morning block of strongly negative coefficients** followed by a progressive return toward the midnight reference in the afternoon and evening. This directly reinforces the raw daily cycle observed in the EDA.

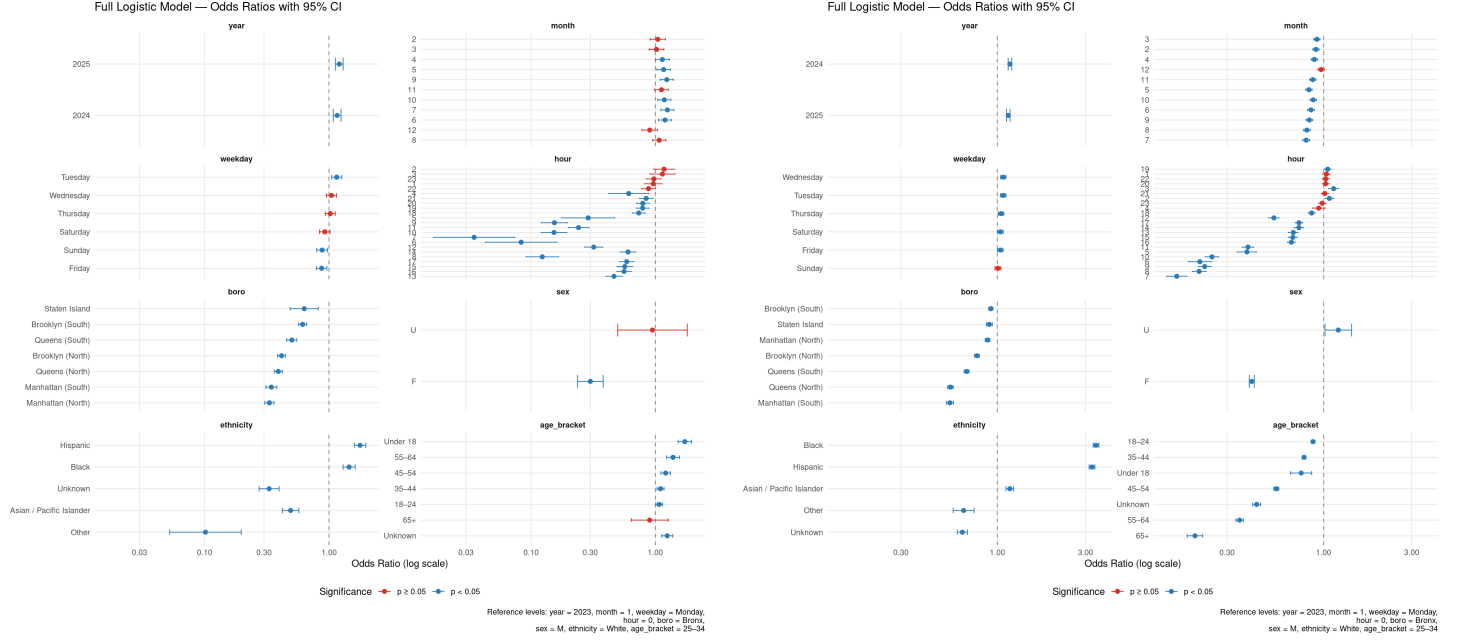
Borough also remains relevant. Using the Bronx as reference, Manhattan North and Manhattan South have odds ratios of approximately 0.332 and 0.344 in the two wheeled model. These correspond to about 66.8% and 65.6% lower conditional odds than the Bronx. The geographical differences are smaller for four wheeled vehicles, where Manhattan South and Queens North are approximately 0.554 and 0.558. This is consistent with Cramer’s V .

Sex is consistent across the two datasets. Female observations have $OR = 0.300$ for two wheeled vehicles and $OR = 0.408$ for four wheeled vehicles relative to Male. In odds terms, this corresponds to approximately 70.0% and 59.2% lower conditional odds of arrest.

Reported ethnicity shows the strongest demographic impact. Relative to White, the two wheeled model estimates $OR_{Black} = 1.451$ and $OR_{Hispanic} = 1.778$. For example, $OR_{Black} = 1.451$ means approximately 45.1% higher estimated odds of arrest than the White reference, conditional on the other included predictors. In the four wheeled model the contrast is much larger, with $OR_{Black} = 3.425$ and

$OR_{Hispanic} = 3.258$. These values correspond to about 3.43 and 3.26 times the conditional odds of the White reference category.

Age is the main point of disagreement. In the two wheeled model, Under 18 has $OR = 1.725$ relative to the 25–34 reference, approximately 72.5% higher conditional odds. In the four wheeled model the same category has $OR = 0.756$, while the older groups decline progressively to $OR_{65+} = 0.201$.



(a) Two wheeled vehicles

(b) Four wheeled vehicles

Figure 2.1: Logistic Regression odds ratios. The main conditional differences reproduce the temporal and demographic patterns identified during the EDA, with substantially different magnitudes across vehicle populations.

LASSO: The penalty is selected by cross validated ROC-AUC. For two wheeled vehicles, $\lambda_{min} = 5.3 \times 10^{-5}$ gives CV-AUC 0.7394, while $\lambda_{1se} = 7.11 \times 10^{-4}$ gives 0.7373. The number of non zero coefficients falls from 62 to 53. For four wheeled vehicles, CV-AUC changes only from 0.7524 to 0.7515 while the non zero coefficients fall from 63 to 50. The **1-SE solution, therefore, achieves a useful reduction in complexity with negligible loss of ranking performance.** More importantly, **shrinkage does not remove the main story.**

In the two wheeled model, the hour 07:00 odds ratio moves from 0.043 at λ_{min} to 0.141 at λ_{1se} , but it **remains one of the largest coefficients.** **Black and Hispanic are almost unchanged**, from 1.452 to 1.473 and from 1.776 to 1.780. Female remains strongly negative and borough also remains relevant. The four wheeled model behaves similarly. At λ_{1se} , $OR_{Black} = 3.202$, $OR_{Hispanic} = 3.057$, $OR_{Female} = 0.420$ and $OR_{hour7} = 0.228$. Older age groups also retain large negative coefficients. The **important interpretation is that the same temporal and demographic axes remain predictive even when weaker dummy coefficients are aggressively shrunk.**

Generalized Additive Models: Logistic Regression and LASSO show that hour and age matter, but their categorical representation is not ideal. Hour is cyclic and the EDA suggests a smooth daily pattern, while age behaves differently across ranges and vehicle types. GAMs address this specific limitation without abandoning interpretability.

GAM, Hour as a Non Linear Daily Cycle: The hour smooth is highly significant in both datasets, with $edf = 7.52$ for two wheeled vehicles and $edf = 7.925$ for four wheeled vehicles, both with $p < 2 \times 10^{-16}$. Effective degrees of freedom above one provide direct evidence that a simple linear effect would be inadequate. More importantly, **the estimated shape is almost the same in both populations**. Conditional arrest propensity is elevated during late night and very early morning hours, declines sharply from about 04:00, reaches a minimum around 07:00 to 08:00, and then increases through the afternoon and evening. The fitted two wheeled smooth spans about 2.3 log odds units between minimum and maximum, roughly a ten fold change in conditional odds. The four wheeled smooth spans about 1.9 log odds units, around 6.7 times in odds. This is the strongest result that consistently survives the EDA, Logistic Regression, LASSO and GAM.

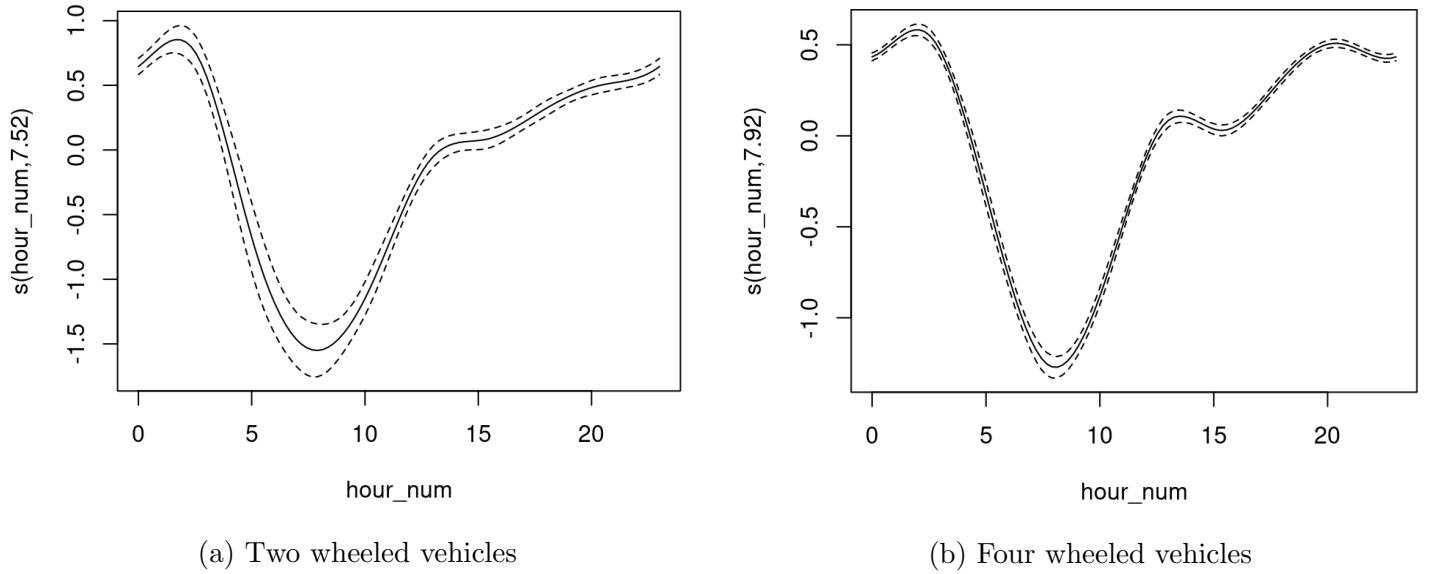
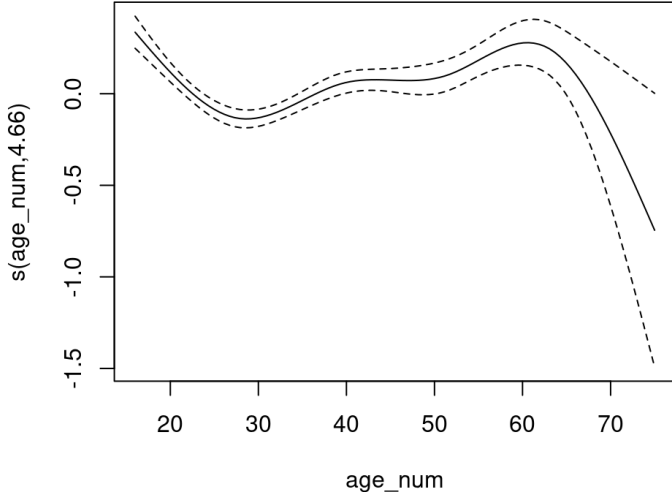
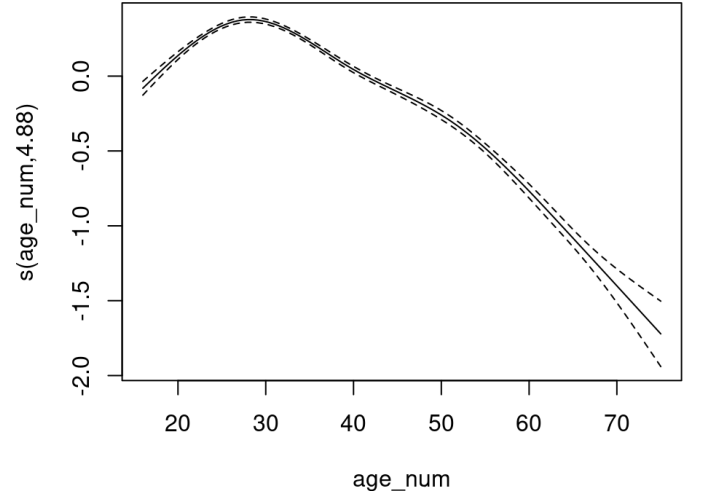


Figure 2.2: GAM partial effects for hour. The daily cycle is strongly non linear and highly consistent across the two datasets.

GAM, Age as a Vehicle Specific Effect Age is also clearly non linear, with $edf = 4.66$ for two wheeled vehicles and $edf = 4.878$ for four wheeled vehicles, again with $p < 2 \times 10^{-16}$. Unlike hour, however, the shapes differ a lot. For two wheeled vehicles, conditional arrest propensity is relatively elevated among the youngest observations, while for four wheeled vehicles, the fitted effect reaches its maximum around ages 27–30.



(a) Two wheeled vehicles



(b) Four wheeled vehicles

Figure 2.3: GAM partial effects for age. In contrast with hour, age follows substantially different conditional patterns in the two vehicle populations.

The new formulation does not change much in the demographic associations. For two wheeled vehicles, the GAM estimates $OR_{Black} = 1.563$, $OR_{Hispanic} = 1.900$ and $OR_{Female} = 0.309$. For four wheeled vehicles the corresponding values are 3.349, 3.185 and 0.403.

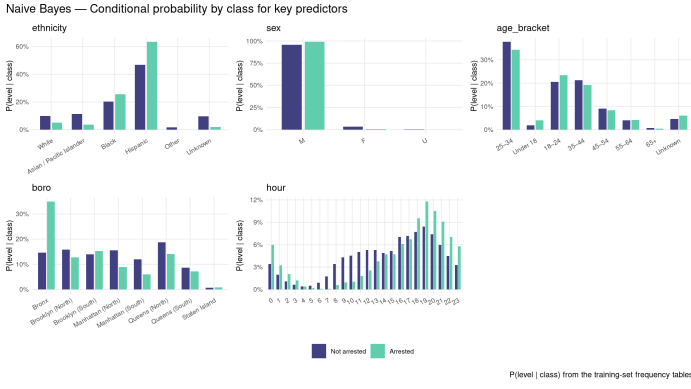
However, the model fit remains limited: the two wheeled GAM explains approximately 9.61% of deviance, while the four wheeled model explains about 9.31%.

Naive Bayes: Naive Bayes provides a different check because it does not rely on the same additive logit formulation. In fact, it models $P(X | Y)$ and combines class conditional distributions through Bayes' rule rather than directly fitting $P(Y | X)$.

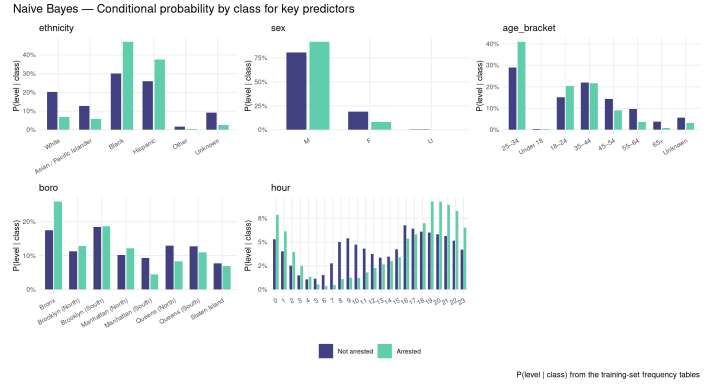
The **class priors immediately reflect the imbalance**, with $P(Y = 1) = 0.0538$ for two wheeled vehicles and 0.0325 for four wheeled vehicles.

Conditional distributions nevertheless reproduce several patterns already observed. In the two wheeled data, Hispanic motorists constitute about 63.4% of arrests and 46.9% of non arrests, while the Bronx accounts for about 34.9% of arrests and 14.6% of non arrests. In the four wheeled data, Black motorists account for about 47.0% of arrests and 30.0% of non arrests, Hispanic motorists for 37.5% and 26.0%, while White motorists show the opposite pattern at about 6.9% and 20.3%.

These are class conditional compositions, not arrest probabilities. Their value is that a method based on a very different modelling assumption still identifies ethnicity, time and geography as useful separators. At the same time, the large overlap between the classes explains why its final discrimination remains only moderate.



(a) Two-wheeled vehicles



(b) Four-wheeled vehicles

Figure 2.4: Selected class-conditional probabilities from Naive Bayes. Ethnicity, hour, and geographical context exhibit visible differences between the arrested and non-arrested classes.

Predictive Comparison No model dominates. Four wheeled outcomes are consistently easier to rank, with ROC-AUC around 0.73 to 0.74 compared with about 0.68 to 0.69 for two wheels. PR-AUC is numerically larger for two wheels because arrest prevalence is also larger. Precision remains low for all models, confirming that these methods are more useful for conditional interpretation than for reliable individual arrest classification.

Model	2W AUC	2W PR	2W Bal.Acc.	4W AUC	4W PR	4W Bal.Acc.
Logistic Regression	0.6868	0.1207	0.633	0.7352	0.0774	0.683
LASSO λ_{1se}	0.6866	0.1203	0.632	0.7340	0.0763	0.682
GAM	0.6897	0.1247	0.637	0.7347	0.0790	0.683
Naive Bayes	0.6818	0.1169	0.629	0.7322	0.0761	0.680

What we can also say is that, LASSO removes weaker coefficients with almost no loss in AUC, while GAM adds useful non linear interpretation without materially increasing discrimination. Naive Bayes reaches a similar performance range despite its very different assumptions.

Research Questions and Conclusions

Are Arrest Outcomes Uniform Across Recorded Stops? **No.** The EDA shows clear heterogeneity across geography, hour and demographics. The Bronx has the highest raw rate in both populations, night and early morning differ strongly from the morning period, Male observations have higher arrest rates than Female observations, and Black and Hispanic categories have higher raw rates than White. The two datasets also differ materially in arrest prevalence and age structure. These patterns are descriptive and do not yet identify independent associations.

Which Disparities Persist After Adjustment? **Several important disparities persist.** Logistic Regression shows that hour, borough, sex, ethnicity and age remain informative after simultaneous adjustment. LASSO then retains the same major signals under shrinkage, while GAM preserves the main sex and ethnicity effects after modelling age and hour more flexibly. The strongest example is ethnicity in four wheeled stops, where Black and Hispanic remain at roughly three times the conditional odds of the White reference across Logistic Regression, LASSO and GAM. Sex is also consistent, with Female conditional odds around 60% lower than Male in the four wheeled models and around 70% lower in the two wheeled models.

Do Two and Four Wheeled Stops Tell the Same Story? **Only partially.** Both datasets reinforce the temporal pattern, lower Female odds and positive Black and Hispanic associations relative to White. However, geography is stronger for two wheels, ethnicity is much stronger for four wheels, and age follows materially different shapes. The comparison therefore supports the original decision to keep the populations separate rather than estimating one common model.

How Predictable Is Arrest? **Moderately as a ranking problem, weakly as an individual classification problem.** All four models perform better than random ranking, but none provides strong separation. The fact that additional flexibility changes AUC only marginally indicates that model form is not the only constraint. The available variables explain only part of the post stop arrest process, which is also consistent with low precision and the modest deviance explained by the GAMs.

Do the Results Establish Excessive or Discriminatory Enforcement? **No causal conclusion can be established from these data alone.** The evidence supports persistent statistical disparities

among recorded stops because several raw differences survive multivariable adjustment, shrinkage and alternative modelling assumptions. However, the dataset omits the reason for the stop, detailed legal and behavioural circumstances, motorists who were never stopped and other possible confounders. The analysis can therefore document structured post stop disparities, but cannot determine whether those disparities are caused by discriminatory or excessive enforcement.

Conclusions: The strongest conclusion supported by the analysis is therefore that **recorded NYPD vehicle stops exhibit persistent demographic, temporal and geographical disparities in post stop arrest outcomes, with meaningful differences between two wheeled and four wheeled vehicle populations.** The evidence does not establish a clear cause-effect connection behind those disparities and cannot, by itself, prove discriminatory or excessive enforcement. Instead, the analysis identifies where statistical differences remain after the observed context is taken into account and clarifies the limits of what these datasets (and model formulations) can support.